

EDUCATION

McGill University, Schulich School of Music

Ph.D. in Music Technology

Sep. 2022 - Present

Montreal, Canada

Supervisors: Prof. Marcelo M. Wanderley and Assoc. Prof. Isabelle Cossette

Shanghai Jiao Tong University, USC-SJTU Institute of Cultural and Creative Industry

M.A. in Cultural & Creative Industry Management

Sep. 2019 - Mar. 2022

Shanghai, China

Advisors: Assoc. Prof. Gus Xia and Asst. Prof. Zhixin Xu

Peking University, National School of Development

Bachelor in Economics, Double Degree

Sep. 2017 - Jun. 2019

Beijing, China

Beijing University of Posts and Telecommunications, School of Digital Media

B.E. in Digital Media Technology

Sep. 2015 - Jun. 2019

Beijing, China

RESEARCH INTERESTS

My research interests center around music and human-computer interaction, with a particular focus on musical haptics and embodied interaction. By researching music, I aim to inspire the broader HCI field with new paradigms of multimodal interaction, demonstrating how sensing, haptics, and AI-driven modeling of embodied musical practices can inform the design of more intuitive, expressive, and socially connected interactive systems.

PEER-REVIEWED PUBLICATIONS

Journal Articles

- [J.1] **Ziyue Piao**, Christian Frisson, Bavo Van Kerrebroeck, and Marcelo M. Wanderley, "Assessing the Impact of Force Feedback in Musical Knobs on Performance and User Experience", *Actuators*, 2024.

Conference Full Papers

- [C.3] **Ziyue Piao**, Yohei Wada, Isabelle Cossette, Marcelo M. Wanderley, and Akira Maezawa, "Exploring Breathing-Music Coupling: Using the *Breathing Mirror* for Somatic Reflection in Piano Performance", *International Conference on New Interfaces for Musical Expression (NIME)*, London, UK, 2026.
- [C.2] Leon Lu, Chase Crispin, **Ziyue Piao**, Aino Eze-Anyanwu, and Audrey Girouard, "Project TapTap: A Longitudinal Study Exploring Non-Verbal Communication through Vibration Signals Between Teachers and Blind or Low Vision Music Learners", *ACM International Conference on Human Factors in Computing Systems (CHI)*, Yokohama, Japan, 2025
- [C.1] **Ziyue Piao**, Marcelo M. Wanderley, and Felipe Verdugo, "MappEMG: Enhancing Music Pedagogy by Mapping Electromyography to Multimodal Feedback", *International Conference on ArtsIT, Interactivity and Game Creation (ArtsIT)*, Campinas, Brazil, 2023.

Conference Short Papers

Note: All types of publications except full papers

- [P6] **Ziyue Piao**, Katsunori Suzuki, Naoto Kojima, Yohei Wada, Isabelle Cossette, Marcelo M. Wanderley, and Akira Maezawa, "GummiSoma: Co-Designing a Carbon-Nanotube Strain-Sensing Garment for Respiratory Tracking in Vocal Performance", *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC)*, Shanghai, China, 2026.
- [P5] **Ziyue Piao**, Isabelle Cossette, and Marcelo M. Wanderley, "Reconciling Kinesthetic Mismatches: A Somatic Alignment Mindset for Musical Body Transformation", *ACM International Conference on Human Factors in Computing Systems (CHI)*, Barcelona, Spain, 2026.
- [P4] **Ziyue Piao** and Kanyu Chen, "Respiratory Musical Interfaces (RMIs) in NIME", *International Conference on New Interfaces for Musical Expression (NIME)*, London, UK, 2026.
- [P3] **Ziyue Piao**, "Exploring Haptics to Augment Kinesthetic Learning in Singing", *ACM International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*, Bordeaux, France, 2025.
- [P2] **Ziyue Piao** and Gus Xia, "Sensing the Breath: A Multimodal Singing Tutoring Interface with Breath Guidance", *International Conference on New Interfaces for Musical Expression (NIME)*, Online, 2022.
- [P1] Yinmiao Li, **Ziyue Piao**, and Gus Xia, "A Wearable Haptic Interface for Breath Guidance in Vocal Training", *International Conference on New Interfaces for Musical Expression (NIME)*, Shanghai, China, 2021.

Yamaha Corporation

Research Collaboration, Mentors: Dr. Akira Maezawa and Yohei Wada

Sep. 2025 - Present

Hamamatsu, Japan and Remote

Breathing Analysis with Sensing Wearables for Piano Performance:

- Led an in-lab study with seven pianists, capturing synchronized piano performance and respiration data through Yamaha GummiStra wearable breathing belts.
- Built an annotated visual interface that layers multimodal breathing and performance signals to streamline note taking and playback.
- Synthesized qualitative interview findings with quantitative breathing and performance metrics to surface design insights.
- Prototyped and iteratively refined the corset with GummiStra sensor to measure thoracic expansion during music performances.

Sony Computer Science Laboratories

Research Intern, Mentors: Dr. Shinichi Furuya, Asst. Prof. Jun Nishida (University of Maryland)

Jun. 2025 - Sep. 2025

Tokyo, Japan

ScapuLearn: Finger Dynamics Estimation through the Scapular Muscle-skeletal Movements:

- Developed a machine-learning pipeline that process shoulder-mounted IMU data to estimate finger force, loudness, keyboard register, and scapular posture during piano pieces.
- Analyzed shoulder's rotation and acceleration features and trained gradient-boosted models to infer finger-level dynamics from minimally intrusive upper-body sensing.
- Validated the approach through multi-stage user studies with twenty pianists, demonstrating how wearable sensing plus ML can support performance analysis and pedagogy.

Input Devices and Music Interaction Lab & Music Performance and Body Lab, McGill University

Research Assistant, Advisors: Prof. Marcelo M. Wanderley and Assoc. Prof. Isabelle Cossette

Sep. 2022 - Present

Montreal, Canada

Challenges and Design Opportunities in Teaching Singing Breath Control:

- Led focus groups with eleven teachers and students to map kinesthetic pain points in teaching breath control and prioritize assistive technology concepts.

Haptic Music Editing Interface for Pitch Modulation:

- Built a web interface that streams TorqueTuner haptic data into a pitch-modulation synthesizer for real-time editing.
- Conducted controlled studies on how various force-feedback modes affect fine-grained vibrato control.
- Analyzed both pitch-contour performance data and qualitative feedback via Likert-scale ratings and interviews.

Creative Interactions Lab, Carleton University

Visiting Researcher, Advisors: Prof. Audrey Girouard, Leon Lu

May. 2024 - Aug. 2024

Ottawa, Canada

Wearable Haptics for Music Accessibility:

- Performed thematic analysis of a ten-week study with twelve music teachers and visually impaired students to document how a wearable vibrotactile tool expands music-accessibility strategies.

Laboratory of Simulation and Movement Modelling, University of Montreal

Research Intern, Advisors: Prof. Felipe Verdugo and Prof. Marcelo M. Wanderley

Sep. 2022 - Aug. 2023

Montreal, Canada

MappEMG: Transmitting Muscle Effort to Vibration for Piano Pedagogy:

- Implemented real-time muscle-effort capture by integrating Delsys and BITalino EMG streams into a single calibration pipeline.
- Developed the MappEMG system to map EMG data to smartphone vibration for teacher-student feedback.
- Led evaluation workshops in five music conservatories to assess its pedagogical impact with more than forty piano learners.

Computer Music Group, Carnegie Mellon University

Research Assistant, Advisor: Prof. Roger B. Dannenberg

Mar. 2021 - Mar. 2022

Remote

Music Computing and Expectation Modeling:

- Investigated predictive models for musical sequences in contemporary music.
- Enhanced expectation-network algorithms by replacing simple counting with gradient-descent-based learning.

Music X Lab, New York University Shanghai

Research Assistant, Advisor: Assoc. Prof. Gus Xia

Mar. 2020 - Jan. 2022

Shanghai, China

Multimodal Vocal Tutoring System:

- Developed an interactive score that plots breathing traces alongside pitch notation so singers can align phrasing with airflow.
- Integrated airflow, chest-expansion, and audio sensors to detect breathing patterns in real time during lessons.
- Prototyped a wearable device to track dynamic breathing movements.

ACHIEVEMENT

Funding and Grants

- **Graduate Mobility Award**, McGill University, \$3,984 CAD. 2026
- **Globalink Research Award**, Mitacs, \$6,000 CAD. 2026
- **Student Award** (Collaborator: Hanwen Zhang), CIRMMT, \$8,000 CAD. 2025
- **TCH Innovation in Haptics Research Grant**, IEEE Technical Committee on Haptics, \$ 2,300 USD. 2024
- **FRQSC Doctoral Scholarship**, Fonds de recherche du Québec, \$ 83,334 CAD (Top 1/21). 2024 - 2028
- **Student Engagement Fund Winner** (Collaborators: Alberto Acquilino, Theodora Nestorova), McGill University, \$1,500 CAD. 2024
- **Inter-centre Research Exchange Funding**, CIRMMT, \$3,000 CAD. 2023
- **Student Award** (Collaborator: Theodora Nestorova), CIRMMT, \$10,000 CAD. 2023

Research Awards

- **ISMIR WIMIR Accomodation & Registration Grants**, International Society for Music Information Retrieval Conference. 2025
- **Best Oral Presentation**, The 2nd International Electronic Conference on Actuator Technology (IECAT). 2024
- **Best Oral Presentation**, COBS Student Colloquium. 2024
- **Graduate Excellence Award**, McGill University. 2022 - 2025
- **Online Conference Award**, CIRMMT. 2023 - 2024
- **Graduate Research Enhancement and Travel (GREAT) Award**, McGill University. 2023, 2026
- **NIME Ableton Scholarship**, International Conference on New Interfaces for Musical Expression. 2022
- **ISMIR Registration Grant**, International Society for Music Information Retrieval Conference. 2021

Other Honors

- **Differential Fee Waiver for International Students**, McGill University. 2022 - 2023
- **Outstanding Graduate**, Shanghai Jiao Tong University (Top 1%). 2022
- **Outstanding Teaching Assistant Award**, Shanghai Jiao Tong University (Top 1%). 2020
- **Excellent Teaching Assistant Award**, Foundational General Education of Shanghai Jiao Tong University. 2020
- **Outstanding Graduate of Beijing**, Beijing University of Posts and Telecommunications (Top 3%). 2019
- **Honorable Prize**, International Mathematical Contest in Modeling. 2018
- **Hengtong Photoelectric Enterprise Scholarship**, Beijing University of Posts and Telecommunications. 2017
- **1st Prize Academic Excellence Award**, Beijing University of Posts and Telecommunications. 2016, 2018

TEACHING

McGill University

Co-instructor with Champ Darabundit

- MUMT 302 New Media Production 1 2025 Winter
- MUMT 302 New Media Production 1 2024 Winter

Guest Lecturer

- MUPP 693 Performance Practice Seminar, *instructor Prof. Fabrice Marandola*
Feel the Music: Haptics in Music Learning 2026 Winter
- MUPD 235 Music as a Profession, *instructor Dr. Jacqueline Woodley*
Musical Haptics and Musicians 2025 Winter
- MUMT 620 Gestural Control of Sound Synthesis, *instructor Prof. Marcelo Wanderley*
Designing a Multimodal Interface for Music Pedagogy 2023 Fall

New York University

Guest Lecturer

- CSCI-SHU 188 Introduction to Computer Music, *instructor Assoc. Prof. Gus Xia*
How to Design a Singing Tutoring Interface 2021 Fall
- CSCI-SHU 188 Introduction to Computer Music, *instructor Assoc. Prof. Gus Xia*
Design a Single Thread System — a Singing Tutoring UI Demo 2020 Fall

Shanghai Jiao Tong University

Teaching Assistant

- Introduction to Computer Music 2021 Winter
- Symphony 2020 Fall
- Chinese and Western Music Theory 2020 Spring

MENTORING

- **Bélangier Nicolas (2026)**, DEC in Health Science, Sainte-Anne College: Capturing Multiple Bio-signal through BITalino in Music Performance
- **Jiayi Liang (2025)**, MS in Innovation Design Engineering, Imperial College London: Vocal Flux: Visualizing your Voice through Muscle Waves
- **Anya Valimahomed (2024)**, BA in Psychology, McGill University: Using Educational Models for Analyzing Kinesthetic Learning Opportunities
- **Gabrielle Seymour (2023-2024)**, BS in Computer Science, McGill University: Audio-Visual Interactive for Ambient Music Generation

ACADEMIC SERVICES

Committee Leadership Roles

◦ NIME 2027 Design Chair	2026 - 2027
◦ ISMIR 2026 Volunteer Chair	2025 - 2026
◦ IEEE Technical Committee on Haptics (TCH) Student Representative	2025 - 2026
◦ ACM TEI 2026 Registration Chair	2025 - 2026
◦ ACM TEI 2026 Associate Chair of Work-In-Progress Track	2025 - 2026
◦ CIRMMT Student Representative	2024 - 2026
◦ CIRMMT Student Coordinator for Research Axis 3	2023 - 2024
◦ Shanghai Jiao Tong University (SJTU) Academic Affair Fellow	2021 - 2022

Academic Reviews

Reviewed 47 submissions total. Years marked with * indicate Special Recognitions for Outstanding Reviews.

Journals

◦ IEEE Transactions on Haptics (ToH), article.	2026
◦ Springer Nature Scientific Reports, article.	2026

Conferences

◦ NIME, Papers and Workshops.	2021–2026
◦ ACM CHI, Papers and Late Breaking Works.	2025*, 2026*
◦ ACM Design Interactive Systems (DIS), Papers.	2025
◦ ACM TEI, Papers and Pictorials.	2025, 2026
◦ ACM Creativity & Cognition (C&C), Papers.	2025
◦ ACM Ubiquitous Computing (UbiComp-ISWC), Demos and Posters.	2024, 2026
◦ EAI ArtsIT, Interactivity & Game Creation (ArtsIT), Papers.	2024

Volunteer

◦ Volunteer of IEEE Haptic Symposium 2026	2026
◦ Volunteer of ISMIR 2024	2024
◦ Senior volunteer of NIME 2021	2021

TALKS AND PRESENTATIONS

• Measuring, Understanding, and Guiding Breath in Musical Expression, <i>Metacreation Lab, Simon Fraser University.</i>	Aug. 2026
• Music Learning with Haptics: Internal Awareness, Motor Control, and Collaborative Metaphor, <i>CIRMMT Haptics Workshop.</i>	Apr. 2026
• The Duality of Breath: From Physiological Spontaneity to Conscious Control in Musical Expression, <i>McGill Doctoral Colloquium.</i>	Apr. 2026
• How does breathing convey expressive and musical dynamics in piano performances, <i>Yamaha Corporation.</i>	Dec. 2025
• Touching the Unseen: Enhancing Music Learning with Wearable Haptics, <i>Yamaha Corporation.</i>	Sep. 2025
• ScapuLearn: Minimally-intrusive Estimation of Finger and Postural Dynamics via Scapular Musculoskeletal Sensing, <i>Sony CSL.</i>	Aug. 2025
• Bridging Kinesthetic Mismatches in Music Learning with Wearable Haptic Devices, <i>COBS Student Colloquium.</i>	May. 2025
• CantusPlay: An Interactive Singing Interface for Practicing Improvised Renaissance Counterpoint, <i>COBS Student Colloquium.</i>	May. 2025
• Exploring on Breathing Challenges and Technology Assisted Solutions, <i>The Voice Foundation Annual Symposium.</i>	May. 2025
• How to write a success CIRMMT student award proposal, <i>CIRMMT Colloquial Colloquium.</i>	Jan. 2025
• Assessing the Impact of Force Feedback in Musical Knobs on Performance and User Experience, <i>IECAT Conference 2024.</i>	Nov. 2024
• SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, <i>PAVA Symposium 2024.</i>	Oct. 2024

• Preliminary Exploration of Kinesthetic Breathing Feedback for Enhanced Vocal Pedagogy, <i>COBS Student Colloquium</i> .	May 2024
• Designing a Multimodal Interface for Music Pedagogy, <i>CIRMMT Workshop with Prof. Karon MacLean</i> .	Nov. 2023
• SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, <i>CIRMMT MMR Re-Open Celebration</i> .	Oct. 2023
• SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, <i>COBS Student Colloquium</i> .	May 2023
• MappEMG – Feeling the muscle activity of classical musicians, <i>CIRMMT Workshop on Piano Knowledge Transfer</i> .	May 2023
• Sensing the Singing Body, <i>CIRMMT Workshop with Pamela Z</i> .	Jan. 2023
• Sensing the Breath: Multimodal Singing Tutoring Interfaces with Breath Guidance, <i>Music + AI Reading Group, Mila</i> .	Oct. 2022
• Multimodal Singing Tutoring Interfaces Design for Breathing Guidance, <i>IDMIL, McGill University</i> .	Sep. 2022

ARTISTIC PERFORMANCES AND ROLES

Interactive Artworks & Residencies

- **Somatic Resonances for Soprano and Ambient Electronics:** *Composer and Vocal Performer* June 2027
Bled Contemporary Music Week, Bled, Slovenia.
- **Dao: A Laptop Orchestra Performance:** *Composer, Director, & Performer* Dec. 2020
Sprout Art Exhibition, Shanghai, China.

Classical Music Performances

- **C.** (F. Poulenc): *Soprano Soloist* May 2025
Living Your Music Seasonal Recital, Montreal, Canada.
- **He Shall Feed His Flock** (G. Handel): *Soprano-Alto Duet* Dec. 2024
Living Your Music Seasonal Recital, Montreal, Canada.
- **Se tu m'ami** (A. Parisotti): *Soprano Soloist* Dec. 2023
Living Your Music Seasonal Recital, Montreal, Canada.
- **ECNU Alma Mater Song** (Arranged): *Vocal Chamber Quartet* Oct. 2021
Featured on the NYU Shanghai Official WeChat Channel, Shanghai, China.
- **Huo Ba Jie, Red Sorghum, & Kung Fu Panda:** *Orchestra Guzheng Player* Dec. 2019
Jiang Xin Concert, SJTU Orchestra, Shanghai, China.

Artistic Leadership and Roles

- **Vocal Artist (Soprano):** Advanced classical training with Dr. Irene M. Feher and Giancarlo Scalia. 2022 - Present
- **Chamber Vocal Artist (Alto):** Performed polyphonic choral works with SJTU Chamber Choir. 2020 - 2021
- **Principal Guzheng Instrumentalist:** Performed with SJTU Chinese-Western Orchestra. 2019 - 2020
- **Artistic Director:** Curated cross-disciplinary stage productions at School of Digital Media, BUPT. 2015 - 2019

SKILLS

- **Programming:** Python*, C++, Javascript*.
- **Audio Production:** Logic Pro*, Audition, PureData*, MAX MSP
- **Prototyping:** Arduino*, Fusion 360, 3D Printing.
- **VR, AR and Game:** Maya*, Processing*, Unity, After Effects.
- **Graphic Design:** Illustrator*, Photoshop*.
- **Toolkits:** scikit, PyTorch.
- **Languages:** Mandarin (native), English (advanced), French (beginner).

Items marked with *: professional proficiency; others: used at least once in projects.

REFERENCES

Prof. Marcelo M. Wanderley	McGill University	marcelo.wanderley@mcgill.ca
Assoc. Prof. Isabelle Cossette	McGill University	isabelle.cossette1@mcgill.ca
Prof. Roger B. Dannenberg	Carnegie Mellon University	rbd@cs.cmu.edu
Assoc. Prof. Gus Xia	MBZUAI	gus.xia@mbzuai.ac.ae
Dr. Akira Maezawa	Yamaha Corporation	akira.maezawa@music.yamaha.com
Dr. Irene M. Feher	Concordia University	irene.feher@concordia.ca

Available upon request.